

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI-590018**

Report on
INDUSTRY INTERNSHIP (21INT82)
“Nalla-Nudi: Technical Dictionary for Students”

Submitted in the partial fulfillment of the requirement for the Internship Program
Bachelor of Engineering
In
CSE (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Submitted by
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2024-25

CERTIFICATE

This is to certify that Tilak Kadlaskar (1VA21CI058), a bonafide student of SAI VIDYA INSTITUTE OF TECHNOLOGY, Bengaluru, carried out the Industry/Research Internship at “Antigravity” in partial fulfillment for the award of Bachelor of Engineering in CSE (Artificial Intelligence & Machine Learning) VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Belagavi during the year 2024-25. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report. The Internship report has been approved as it satisfies the academic requirements in respect of Internship prescribed for the said degree.

Signature of Guide
Signature of Principal

Signature of HOD

ACKNOWLEDGEMENT

The completion of internship brings with the sense of satisfaction, but it is never completed without thanking the persons who are all responsible for its successful completion.

I wish to express our deep sincere feelings of gratitude to my Institution, Sai Vidya Institute of Technology, for providing me an opportunity to do our education.

I extend my deep sense of sincere gratitude to the Principal, Sai Vidya Institute of Technology, Bengaluru, for having permitted to carry out the internship work.

I express my heartfelt sincere gratitude to the HOD, Department of CSE (Artificial Intelligence & Machine Learning), for valuable suggestions and support.

I am thankful to Antigravity for providing me an opportunity to be an internee and develop the Nalla-Nudi Application.

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ABSTRACT

This report presents a comprehensive overview of the internship experience at Antigravity. The internship encompassed technical training in Android App Development using Kotlin, Room Database, Coroutines, MVVM architecture, and Artificial Intelligence (ML Kit Offline Translation). The primary project was 'Nalla-Nudi', an Android application designed to bridge the language gap for students transitioning from Kannada-medium to English-medium education.

Through hands-on coding, UI/UX design, and AI integration, the internship provided valuable insights into the development of production-ready mobile applications. This document outlines the project's background, technologies used, tasks undertaken, and the technical competencies developed during the internship period.

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Nalla-Nudi (ನಲ್ಲ ನುಡಿ) is a premium, Material Design 3-based Android application designed to bridge the language gap for students transitioning from Kannada-medium to English-medium education. The transition often introduces significant challenges in understanding technical terminology across various subjects. Nalla-Nudi addresses this by providing instant, offline translations and explanations of technical terms in both English and Kannada.

1.2 OBJECTIVES

1. To design and develop a user-friendly Android application with a modern UI (Material Design 3).
2. To implement an efficient offline dictionary using Room Database for quick term retrieval.
3. To integrate AI-powered translation (Google ML Kit) for words not present in the offline library.
4. To provide an accessible experience through Text-To-Speech (TTS) for correct pronunciation.
5. To implement features like Subject-Wise Filtering, Favorites, Search History, and Word of the Day.

1.3 LEARNING ACTIVITIES

The learning activities included comprehensive training in Android development with Kotlin. This involved understanding the MVVM architecture, utilizing Kotlin Coroutines for asynchronous programming, and designing responsive layouts using XML and Material Design guidelines. Further activities included integrating third-party APIs and machine learning models directly onto the mobile device for offline capabilities.

1.4 JOB DESCRIPTION

During the internship, the primary role was working as an Android Application Developer for the Nalla-Nudi project. Responsibilities included developing the frontend interface, setting up the local Room Database, seeding the database with a JSON file, integrating Text-to-Speech, and incorporating Google ML Kit for AI translations. Regular debugging and UI/UX optimization were also part of the daily tasks.

CHAPTER 2: PROJECT OVERVIEW AND LEARNING OUTCOME

2.1 OVERVIEW OF THE WORK CARRIED OUT

The development of Nalla-Nudi progressed through several stages:

1. UI/UX Design: Created wireframes and implemented Material Design 3 components, including dynamic theming (Dark/Light modes).
2. Database Integration: Configured Room Database to handle a vast dictionary of technical terms across subjects like Physics, Biology, Math, Science, and Commerce.
3. Core Features: Implemented lightning fast offline search, subject filtering, and a 'Word of the Day' feature.
4. AI & Accessibility: Integrated ML Kit for offline AI translations and Text-To-Speech for voice guidance.

2.2 TOPICS LEARNED DURING THE INTERNSHIP

- Kotlin Programming Language
- MVVM Architecture
- Android Room Persistence Library
- Kotlin Coroutines and Flow
- Material Design 3 (M3)
- Google ML Kit (Offline Translate)
- Text-To-Speech (TTS) Integration

2.3 PROJECT DEVELOPED / RESULT DISCUSSION

The final product, Nalla-Nudi, successfully met all objectives. The application is highly responsive, visually appealing, and provides accurate technical translations. The combination of an offline database with an AI fallback ensures that students always get an answer. The application is currently published and available for download, with its source code managed efficiently.

2.4 IMPORTANCE AND APPLICATIONS

Nalla-Nudi holds immense importance for rural and regional-medium students in Karnataka. It democratizes technical education by removing language barriers, making complex subjects more approachable. The offline capability ensures it is useful even in areas with poor internet connectivity.

CHAPTER 3: CONCLUSION AND REFERENCES

3.1 CONCLUSION

The internship at Antigravity was highly productive, resulting in the successful development of the Nalla-Nudi application. The experience provided deep technical knowledge in modern Android app development, from architecture to AI integration. The project not only enhanced technical skills but also contributed to a meaningful educational tool that addresses a real-world problem.

3.2 REFERENCES

- [1] Android Developers Documentation: <https://developer.android.com/>
- [2] Kotlin Language Documentation: <https://kotlinlang.org/>
- [3] Google ML Kit Documentation: <https://developers.google.com/ml-kit>
- [4] Material Design 3 Guidelines: <https://m3.material.io/>

APPENDIX: DAILY ACTIVITY LOG

Week 1: Setup Android Studio, learnt basic Kotlin syntax, and designed the initial UI mockups.

Week 2: Studied MVVM architecture and implemented the Room Database with entity definitions for Technical Terms.

Week 3: Seeded the database using JSON assets and implemented the offline search functionality using Coroutines and Flow.

Week 4: Designed Material 3 UI components, including the chip group for subject filtering and RecyclerView for search results.

Week 5: Implemented the 'Word of the Day' feature and integrated Text-To-Speech (TTS) for English and Kannada.

Week 6: Integrated Google ML Kit for offline AI translations to act as a fallback for missing terms.

Week 7: Handled Dark/Light mode theming and user preferences (SharedPreferences) for search history and favorites.

Week 8: Conducted extensive testing, debugging, UI polishing, and prepared the final project report.